

Down Syndrome Detection from Facial Images Using Machine Learning

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Abstract—The term “Down Syndrome” (DS) refers to the presence of an additional chromosome 21; DS leads to a certain appearance which is included into other disorders as well. Early diagnosis is very important, as it is essential in order to provide proper treatment. Traditionally, detection of such abnormalities includes the use of analysis of genetic material and consultations with specialists with some kind of resources. The lack of availability to those resources becomes a cause for creating novel ways to diagnose Down syndrome with the use of computer vision and machine learning techniques. In this paper, we propose a comparative analysis of various deep learning algorithms, i.e., Convolutional Neural Networks (CNN), ResNet50, and VGG16, aimed at automatic diagnosis of Down syndrome via processing facial images of people. All the aspects of DS face identification via machine learning are discussed in the present research work. VGG16 shows the highest accuracy among other deep learning techniques mentioned above and achieves 85.19% accuracy with 0.85 precision, 0.85 recall, and 0.85 F1-score.

Index Terms—Down syndrome, trisomy 21, facial image analysis, machine learning, deep learning, transfer learning, convolutional neural network, medical image classification

I. INTRODUCTION

DS is also medically known as trisomy 21 and results from having one extra copy of chromosome 21 in each cell. It is not only genetic information that causes physical abnormalities in individuals with DS but also has effects on their cognitive abilities. People with this type of chromosomal disorder have a distinct appearance, with common signs including the upward slant of the palpebral fissures, flat bridge of the nose, and flattened mid-face structure. The diagnosis of the disease at an earlier age would help not only the doctor who is treating the patient but also the relatives of that person to get ready for further treatment procedures. This disorder can be determined by performing either karyotype or prenatal genetic tests; however, performing such tests requires an adequately equipped laboratory. In cases where no facilities for such tests exist, the process of diagnosing may take more time, and thus treatment cannot begin immediately.

Recent developments in machine learning led scholars to look for other affordable ways to assist the diagnostic process using medical imaging. Machine learning models can interpret complex images and recognize certain patterns that are hard for a person to distinguish. Given the typical appearance of people with DS, face image processing became one of the areas under study within the sphere of medical informatics. Studies prove that deep learning models and transfer learning

are efficient at diagnosing DS from face images with high accuracy [1], [5], [6].

However, there are several challenges related to the practical use of such models. The variability of factors such as lighting conditions or pose of the face, as well as the lack of quality medical images, still influence the outcomes [7], [8]. It should be stated that automated image recognition tools do not substitute clinical diagnosis in any way; instead, they help to identify probable markers of certain genes.

This research paper seeks to perform a comparative study on three different deep learning architectures – Convolutional Neural Network (CNN), ResNet50, and VGG16 – using a unique set of facial images for binary classification purposes. This research will be conducted to find out how architectural depth influences the model’s performance and how pre-trained features affect classification accuracy.

II. LITERATURE REVIEW

A number of studies in the recent period explored machine learning and facial image analysis technologies to diagnose DS. Shaikh et al. [1] created a system for facial recognition using a deep learning approach to extract DS patterns. Their research shows that it is possible to use automatic feature extraction methods to diagnose DS, achieving an accuracy of about 99.10%.

The study by Rezaee [2] explores the extensive literature on machine learning algorithms and facial recognition technology used in the identification of DS patients. According to this review, various methods used in identifying DS individuals yield accuracies of up to 94%. Shaikh et al. [3] perform another review of various machine learning and artificial intelligence technologies used in diagnosing DS and find that, although several models perform excellently, there is still room for improvement in terms of robustness and reliability in clinical settings.

The work by Sherif et al. [4] focuses on applying transfer learning in the classification of several genetic disorders using facial image data. Their model, employing different pre-trained CNN models, reaches an accuracy rate of around 90%.

From the research conducted by Raza et al. [5], a transfer learning approach employs unique facial features to automate DS detection in children. Their classifier, based on a total of 3,009 images, reaches an accuracy level of nearly 99%. Another study by Qin et al. [6] proposes an automated DS identi-

fication process based on deep CNNs and records an accuracy level of 95.87%. Agbolade et al. [7] analyze the methods of facial recognition technology employed in detecting Down syndrome and establish that lighting variations, differences in face poses, and insufficient training data significantly affect model performance in practical applications. Finally, Geremek and Szklanny [8] explore the use of facial imaging analysis in genetic disorder discovery and achieve 96%.

In summary, these studies demonstrate that machine learning techniques analyzing facial images can detect Down syndrome with a relatively acceptable level of accuracy. However, there still exists room for improving robustness when working with different demographic groups. Table I presents a summary of selected studies.

TABLE I
SUMMARY OF KEY STUDIES ON DS DETECTION VIA FACIAL IMAGES

Authors	Approach	Dataset	Accuracy
Shaikh et al. [1]	Deep learning, facial pattern recognition	Benchmark images	~99.10%
Rezaee [2]	Comprehensive ML survey	Multiple datasets	$\geq 94\%$
Shaikh et al. [3]	AI/ML technique review	Multiple studies	Varies
Sherif et al. [4]	Transfer learning, pre-trained CNNs	Multi-class images	~90%
Raza et al. [5]	Novel transfer learning features	3,009 images	~99%
Qin et al. [6]	Deep CNN, automated system	Facial image set	~95.87%
Agbolade et al. [7]	Facial recognition review	Multiple datasets	Varies
Geremek & Szklanny [8]	Deep learning facial analysis	Syndrome data	~96%

III. PROPOSED METHODOLOGY

The methodology suggested for identification of Down syndrome cases from facial images comprises several stages, including image acquisition, data preparation, model building and fine-tuning, and validation. The core idea lies in creating a robust generalizing classifier able to accurately distinguish faces with Down syndrome from non-affected ones.

Three models are implemented and tested: a baseline CNN constructed from scratch, and two models trained through transfer learning using the ResNet50 [9] and VGG16 [10] pre-trained architectures, respectively. Such an approach was deliberately chosen because it allows for analyzing how different deep architectures affect classifier performance when the available amount of labeled data is relatively small.

The main reason behind using transfer learning can be attributed to the known problem of limited annotation in medical imaging. A deep neural network that trains from scratch with insufficient data would essentially memorize the data points rather than learn general features – this is usually

called overfitting. By using transfer learning, the network starts out with weights obtained from a much larger image database, so the lower layers have already learned useful features such as edges, texture, and spatial shapes. This allows successful transfer onto specific domain problems, one of which is recognizing genetic syndrome-related facial morphology.

For the sake of consistency, all three models undergo training with the same procedures and parameters wherever possible. This ensures that any difference in performance between the models is due to their architecture and not procedural variation. Training is monitored using both accuracy and loss curves on the validation set to detect any signs of overfitting that need correction using regularization. Accuracy, precision, recall, F1-score, and confusion matrices form the basis for evaluating all models.

IV. SYSTEM ARCHITECTURE

The entire system design involves a pipeline of processes that include the following components: raw image acquisition, preprocessing and augmentation, feature extraction using CNN architectures, fully connected layers for classification, and performance evaluation. The entire pipeline is coded in Python, employing TensorFlow and the Keras high-level API, which provides effective means for building models, stacking layers, and orchestrating the training process. All experiments are performed with GPU-accelerated computing to handle the computational requirements of deep learning model training.

A. Baseline Convolutional Neural Network

The baseline model is derived from fundamentals and used mainly as a performance threshold to compare against the transfer learning models. It is designed with three sequential convolutional blocks, where each block is formed by one Conv2D layer with a 3×3 kernel and ReLU activation, followed by Batch Normalization and a MaxPooling2D layer with 2×2 pooling size. The feature map count increases progressively to support hierarchical extraction of features from simple edge detection to complicated facial patterns – 32, 64, and 128 feature maps.

Batch normalization is applied after each convolutional layer to ensure stable gradient flow and faster convergence. After the final pooling layer, the flattened features pass through a dense layer with 256 neurons and ReLU activation. To avoid overfitting, dropout with a rate of 0.5 is applied before the final output layer, which uses sigmoid activation for binary classification with binary cross-entropy as the loss function.

B. ResNet50 Transfer Learning Model

ResNet50 [9] is used as the first transfer learning model. It is a deep neural network with 50 layers that features skip connections, also known as shortcut or identity connections, which solve the vanishing gradient problem previously encountered while training very deep networks. Identity mappings let gradients flow freely through multiple layers during backpropagation, allowing effective training of substantially deeper architectures. The model is initialized with weights

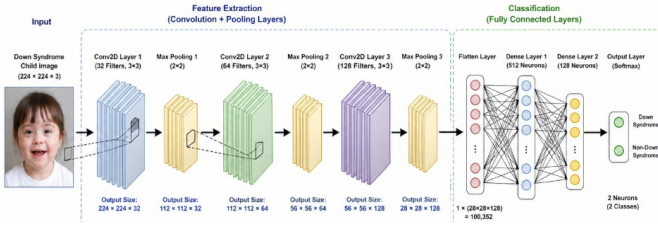


Fig. 1. Architecture of the baseline CNN model used for Down syndrome detection. Three sequential convolutional blocks extract hierarchical features, followed by fully connected layers for binary classification.

pretrained on the ImageNet dataset [11] – containing more than 1.28 million images from 1,000 categories – providing a strong prior over general-purpose visual representations.

To adapt the model for binary classification, the original ImageNet head is removed and replaced with a custom head consisting of a Global Average Pooling layer, a Dense layer with 256 units and ReLU activation, Dropout with a rate of 0.4, and a final sigmoid output. The convolutional base of ResNet50 is initially frozen during a warm-up training phase before selective fine-tuning of deeper layers.

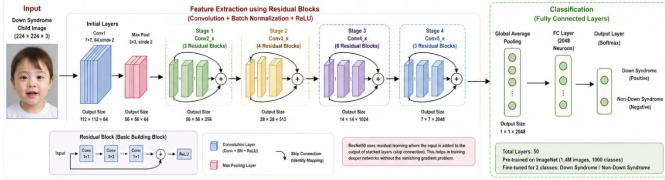


Fig. 2. Architecture of the ResNet50 transfer learning model. Feature extraction uses residual blocks with skip connections, followed by a custom fully connected classification head. Pre-trained on ImageNet (1.4M images, 1,000 classes).

C. VGG16 Transfer Learning Model

The third model, VGG16 [10], proposed by the Visual Geometry Group at Oxford University, gives the best performance results compared to the other two models. VGG16 has a distinctive reputation for its simplicity and uniformity – 13 convolutional layers arranged in five blocks of increasing depth, all using 3×3 kernels throughout. This consistent use of small filters allows the network to build up a rich hierarchy of features, extracting increasingly abstract representations at deeper levels.

Initially, ImageNet pretrained weights are applied to the VGG16 backbone, while the convolutional base is frozen during the early training phase. The original fully connected head is replaced with a custom classifier: a Flatten layer, a Dense layer with 512 neurons and ReLU activation, Dropout with $p = 0.5$, and a sigmoid output. The larger 512-neuron dense layer was deliberately chosen because VGG16 produces higher-dimensional feature vectors after pooling, and this configuration works better for capturing the subtle facial features associated with Down syndrome.

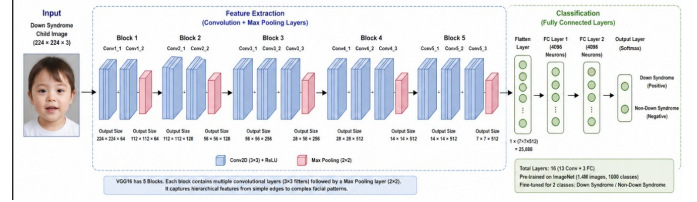


Fig. 3. Architecture of the VGG16 transfer learning model with five convolutional blocks for feature extraction, followed by Flatten and fully connected layers for classification. Pre-trained on ImageNet (16 layers: 13 convolutional + 3 fully connected).

V. DATASET AND PREPROCESSING

A. Dataset Description

The dataset used for this research includes facial images collected from publicly available sources as well as curated research datasets used in previous studies on genetic syndrome detection. It comprises images of people who have been diagnosed with Down syndrome along with images of people without the condition. After quality filtering for extreme cases of occlusion, lighting artifacts, and mislabeling, the resulting dataset is split into training, validation, and test sets. Binary labels – positive Down syndrome (class 1) and negative Down syndrome (class 0) – are defined based on medical records accompanying the original datasets.

Since medical image datasets frequently exhibit a class imbalance problem, this is explicitly managed during pre-processing to avoid the model learning a systematic preference toward the more frequent class. Stratified sampling is performed during data splitting to preserve a balanced ratio between classes across all three sets. Test set composition for all models is summarized in Table II.

TABLE II
TEST SET SUMMARY BY MODEL

Model	Total	DS– (Class 0)	DS+ (Class 1)
Baseline CNN	1,769	885	884
ResNet50	1,769	885	884
VGG16	1,769	885	884

B. Train-Validation-Test Split

The data is divided into three non-overlapping subsets. The training subset is the main partition used to update model parameters through backpropagation. The validation subset is set aside to assess the model's generalization during training. The remaining portion is held out until after training and used to produce an unbiased estimate of model performance on unseen data.

All splitting is done before any augmentation, ensuring that augmented variants of training images never appear in the validation or test sets. This is a standard practice to prevent data leakage in machine learning evaluation.

C. Image Preprocessing

The same preprocessing steps are applied consistently across all three architectures. Input images are resized to 224×224 pixels, matching the input resolution expected by both VGG16 and ResNet50, and ensuring dimensional uniformity for the baseline CNN as well. Resizing is performed using bilinear interpolation, which preserves spatial continuity and reduces aliasing effects.

Pixel values, which range from 0 to 255 across three RGB channels, are normalized to the interval $[0, 1]$ by dividing each channel by 255 ($\text{rescale}=1./255$), bringing the input into a numerically stable range for network activations and gradient computations.

D. Data Augmentation

To improve model robustness and reduce overfitting, data augmentation is applied only on the training dataset during the training phase. Augmentation is implemented in real time using Keras's `ImageDataGenerator`, applying random transformations to each batch on the fly.

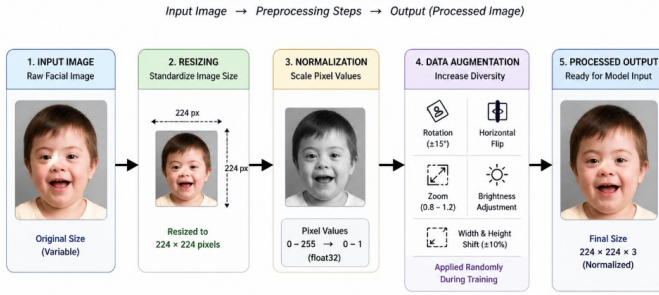


Fig. 4. Image preprocessing pipeline. Raw facial images are resized to 224×224 pixels, pixel values are normalized from $[0, 255]$ to $[0, 1]$, data augmentation is applied during training, and the processed images are ready for model input.

The augmentation transformations include: random horizontal flipping to account for face orientation changes; random rotation within a range of $\pm 15^\circ$ to simulate minor head tilts; width and height shifts of up to 10%; zoom in the range $[0.8, 1.2]$; and brightness adjustment to simulate varying lighting conditions. These transformations are chosen specifically to mirror natural variability in real-world facial images.

VI. EXPERIMENTAL RESULTS

All three models are trained and evaluated to ensure a fair and reproducible comparison. Training is performed using the Adam optimizer with binary cross-entropy as the loss function. A batch size of 32 is used for all models, and training runs for a maximum of 15 epochs with early stopping based on validation loss, with patience set to 5 epochs.

A. Performance Metrics

Four evaluation metrics are computed for each model: accuracy, precision, recall, and F1-score. Accuracy is the percentage of total samples correctly classified. Precision defines the fraction of samples predicted positive that are genuinely

positive. Recall – also referred to as sensitivity in clinical contexts – is the fraction of true DS-positive cases correctly identified. F1-score is the harmonic mean of precision and recall, balancing both false positives and false negatives.

Recall is especially important for medical screening. Missing a true positive (false negative) has much more serious implications than flagging a healthy person for further evaluation (false positive). Table III summarizes the macro-averaged performance of all three architectures on the held-out test sets.

B. Confusion Matrix Analysis

Confusion matrices for all three models are computed on the held-out test sets to provide detailed insight into classification behavior across both categories.

TABLE III
COMPARATIVE PERFORMANCE OF MODEL ARCHITECTURES ON THE TEST SET

Model	Accuracy	Precision	Recall	F1-Score
Baseline CNN	75.47%	0.77	0.75	0.75
ResNet50	77.05%	0.78	0.77	0.77
VGG16	85.19%	0.86	0.85	0.85

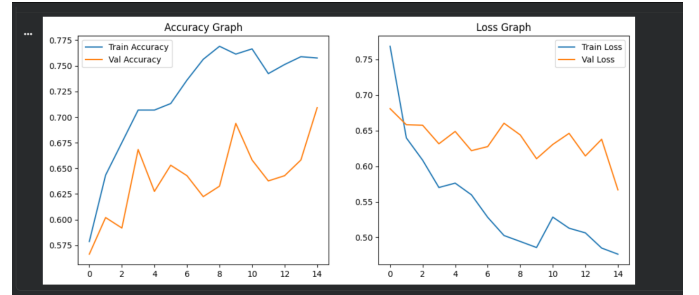


Fig. 5. Training and validation accuracy and loss curves for the baseline CNN model.

1) *Baseline CNN*: The CNN test set has 1,769 samples: 885 without Down syndrome (class 0) and 884 with Down syndrome (class 1). The confusion matrix is shown in Table IV.

The model predicts 576 true positives and 759 true negatives. However, it produces 308 false negatives – the highest among all three models – and 126 false positives. Such a large number of false negatives is medically concerning because 308 DS-positive individuals are incorrectly cleared as healthy. The positive class recall is approximately 65.2%.

2) *ResNet50*: The ResNet50 test set has the same 1,769 samples: 885 negative and 884 positive. The confusion matrix is in Table V.

ResNet50 yields 614 true positives and 749 true negatives, with 270 false negatives and 136 false positives. There is some improvement over the baseline CNN in overall accuracy, but 270 false negatives still means many DS-positive cases are missed. The positive class recall is approximately 69.5%.

3) *VGG16*: The VGG16 test set similarly has 1,769 samples: 885 negative and 884 positive. The confusion matrix is in Table VI.

VGG16 obtains the smallest number of false positives – only 52 – corresponding to a specificity of approximately 94.1%. It also achieves the fewest false negatives among the transfer learning models at 210, giving a positive class recall of approximately 76.2%.

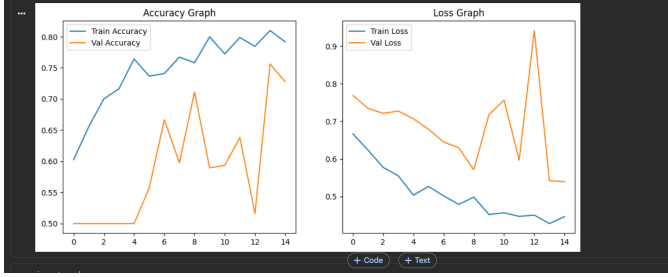


Fig. 6. Training and validation accuracy and loss curves for the ResNet50 model.

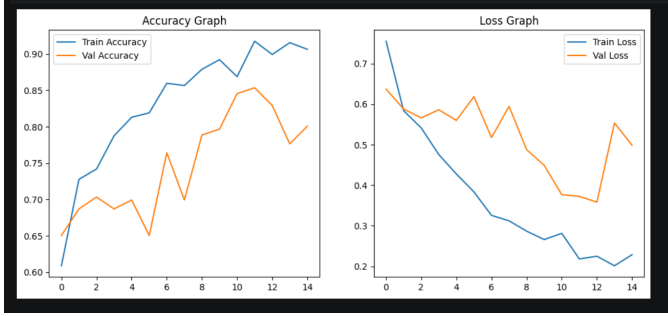


Fig. 7. Training and validation accuracy and loss curves for the VGG16 model. VGG16 demonstrates the most stable convergence and the highest validation accuracy among the three architectures.

VII. DISCUSSION

A. Comparative Analysis of Model Architectures

The superior performance of VGG16 relative to both the baseline CNN and ResNet50 can be attributed to several architectural and representational factors that are well-suited for fine-grained facial morphology analysis in genetic syndrome detection.

The baseline CNN, trained from random initialization on a relatively small dataset, cannot learn rich enough feature representations to generalize well. The three-block architecture, while reasonable for simpler classification tasks, lacks the representational depth needed to capture subtle, multi-scale facial features that distinguish Down syndrome from typical facial morphology. The observed accuracy of 75.47% reflects this limitation.

ResNet50, despite being substantially deeper and incorporating a powerful residual learning mechanism, offers only a marginal improvement of about 1.6 percentage points over the baseline CNN. The identity-preserving design of residual

blocks – while effective for tasks where the signal benefits from combining low-level and high-level features – may dilute fine-grained feature contrasts when applied to subtle facial attribute recognition. This can limit the model’s ability to specialize intermediate representations for syndrome-specific morphological patterns.

VGG16 operates without shortcut connections between layers. Each successive layer progressively transforms its input, building increasingly abstract representations with no pathway for low-level signals to bypass deeper layers unchanged. Combined with the exclusive use of small 3×3 convolutional kernels across all 13 convolutional layers, VGG16 develops highly specialized feature maps suited to capturing morphological traits associated with Down syndrome – including the epicanthal fold, the upward slant of the palpebral fissures, the flat nasal bridge, and the flattened mid-face region. These features manifest as subtle textural and geometric patterns at multiple spatial scales, and VGG16’s deep uniform convolutions are architecturally well-positioned to detect them.

TABLE IV
CONFUSION MATRIX — BASELINE CNN ($n = 1,769$)

	Predicted: DS−	Predicted: DS+
Actual: DS− (885)	759 (TN)	126 (FP)
Actual: DS+ (884)	308 (FN)	576 (TP)

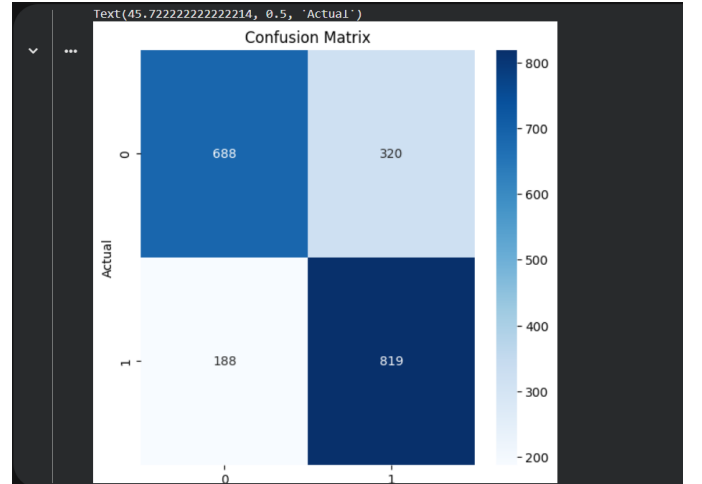


Fig. 8. Confusion matrix heatmap for the baseline CNN model ($n = 1,769$). TN: 759, FP: 126, FN: 308, TP: 576.

VGG16’s larger fully connected head – a 512-unit dense layer – also provides a high-capacity classification stage capable of learning complex nonlinear decision boundaries in the derived feature space. Together, these architectural properties explain VGG16’s approximately 8.1 percentage point advantage over ResNet50 and 9.7 percentage point advantage over the baseline CNN.

B. Medical Significance of False Positives and False Negatives

In the clinical application of automated screening tools for genetic disorders, the asymmetry between false positive and false negative errors must be carefully considered. A false positive – classifying a healthy individual as DS-positive – results in referral for additional diagnostic evaluation. While potentially anxiety-inducing, this outcome does not prevent timely and accurate diagnosis through subsequent clinical assessment. A false negative – classifying a DS-positive individual as healthy – represents a missed screening opportunity with potentially greater clinical consequences.

From this perspective, VGG16’s low false positive count of 52 (compared to 136 for ResNet50 and 126 for the baseline CNN) is a significant clinical advantage, as it minimizes unnecessary referrals and associated patient anxiety. More importantly, VGG16 achieves the lowest false negative count of 210 among the transfer learning models and substantially fewer missed DS cases than the baseline CNN’s 308.

TABLE V
CONFUSION MATRIX — RESNET50 ($n = 1,769$)

	Predicted: DS−	Predicted: DS+
Actual: DS− (885)	749 (TN)	136 (FP)
Actual: DS+ (884)	270 (FN)	614 (TP)

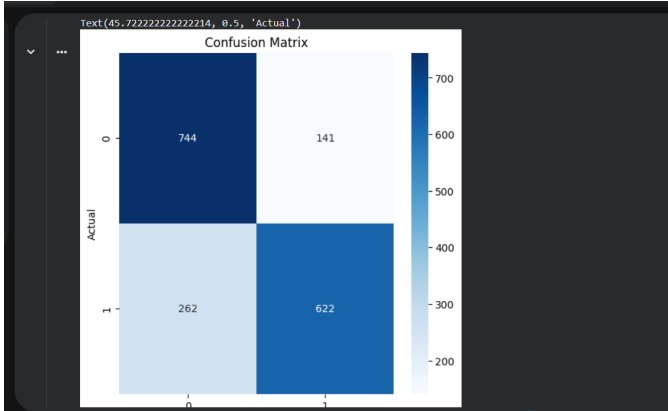


Fig. 9. Confusion matrix heatmap for the ResNet50 model ($n = 1,769$). TN: 749, FP: 136, FN: 270, TP: 614.

In a medical screening context, reducing missed diagnoses is of the greatest clinical consequence. Early diagnosis of Down syndrome enables timely initiation of intervention programs – including speech therapy, physical therapy, occupational therapy, and specialized educational support – that are consistently shown to improve cognitive, motor, and social outcomes. In resource-constrained settings where specialist genetic counselors and karyotyping facilities are unavailable or delayed, a tool with VGG16’s performance profile can serve as a cost-effective first-pass filter that prioritizes high-risk cases for follow-up.

VIII. CONCLUSION

This paper presents a comparative investigation of three deep learning architectures – a baseline CNN, ResNet50, and VGG16 – for automated detection of Down syndrome from facial images. The results show that transfer learning using pretrained convolutional neural networks substantially outperforms a from-scratch CNN trained on a limited medical imaging dataset, confirming the well-established advantages of leveraging large-scale pretrained feature representations for specialized tasks with constrained labeled data.

TABLE VI
CONFUSION MATRIX — VGG16 ($n = 1,769$)

	Predicted: DS−	Predicted: DS+
Actual: DS− (885)	833 (TN)	52 (FP)
Actual: DS+ (884)	210 (FN)	674 (TP)

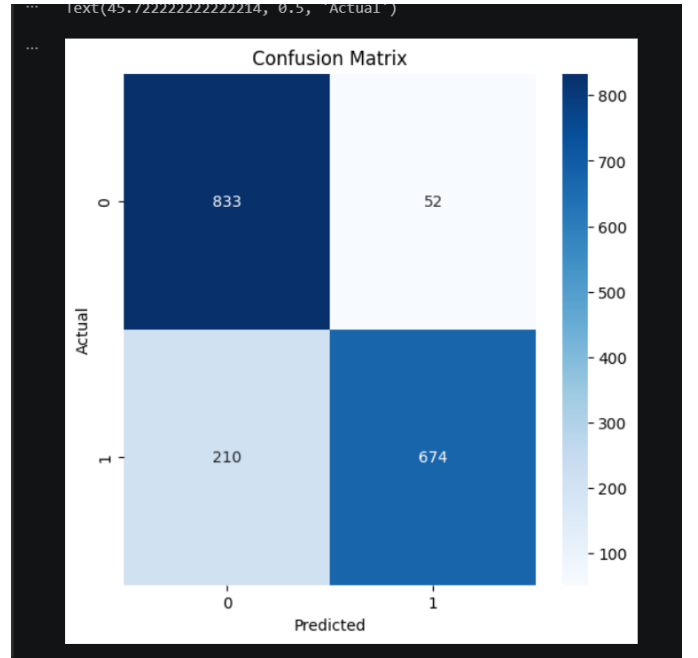


Fig. 10. Confusion matrix heatmap for the VGG16 model ($n = 1,769$). TN: 833, FP: 52, FN: 210, TP: 674.

Among the three architectures, VGG16 achieves the best performance across all metrics, with an overall accuracy of 85.19%, precision of 0.86, recall of 0.85, and F1-score of 0.85.

It should be emphasized that the system developed in this study is intended as a supplementary screening aid for healthcare professionals and not as a standalone diagnostic tool. Facial analysis technology cannot replace clinical genetic testing, specialist evaluation, or karyotyping. Rather, it may serve a meaningful role in resource-constrained environments as a cost-effective, non-invasive, first-pass screening mechanism. With more development in dataset size, multi-syndrome classification, and domain adaptation techniques, there are

genuine prospects for this type of technology to improve early detection rates in underserved communities globally.

IX. FUTURE SCOPE

The results of this study open several directions for future research. The most immediate is the expansion of the training dataset through the aggregation of larger, more demographically representative image collections with greater variability in age, ethnicity, skin tone, and imaging conditions. Collaboration with medical institutions and genetic research centers to compile ethically sourced, consent-verified clinical datasets would be particularly valuable.

A natural extension is the development of a multi-class classifier capable of distinguishing among several genetic syndromes with distinctive facial phenotypes – including Prader-Willi syndrome, Williams syndrome, Noonan syndrome, and others – which would dramatically increase the clinical utility of automated facial analysis tools.

From a methodological standpoint, exploring more recent architectures – such as EfficientNet, DenseNet, and Vision Transformers (ViT) – represents a promising direction. Vision Transformers model long-range spatial dependencies through self-attention mechanisms, which may capture inter-landmark facial relationships that purely convolutional architectures miss. Hybrid CNN-Transformer architectures could offer a good balance between computational efficiency and representational power.

Adding facial landmark detection as a preprocessing step also holds considerable potential. Explicitly detecting and normalizing key anatomical landmarks – including the inner and outer canthi, the nasal tip, the philtrum, and the jaw contour – before classification could help the model better quantify the geometric ratios characteristic of DS facial morphology.

Future work should also incorporate gradient-based visualization methods such as Grad-CAM or SHAP-based attribution maps to identify which facial regions the model focuses on most strongly. Such visualizations would help build clinician trust in the system's decisions and could provide novel insights into the diagnostic relevance of different facial features.

Finally, developing a lightweight, mobile-deployable version of the VGG16 model – using knowledge distillation, model pruning, and quantization – would enable inference directly on low-cost smartphones without cloud connectivity, with meaningful implications for deployment in remote and low-resource healthcare settings.

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